Lab#03

Construct of Escape sequence, variables, various data types and in java.

Objective

To get familiarity with the escape sequence, variables and various data types in java.

Lab Goals

To provide working insight into the usage of various java programming tools:

a. Demonstrate the working of class.

b. Understand escape sequences (\n, \t, \", \\, etc.) in Java for manipulating output.

c. Learn naming conventions for identifiers in Java.

d. Declare and initialize variables of different data types.

e. Understand numerical data types (int, long, short, byte) and their ranges.

f. Explore floating-point data types (float, double) and their precision.

g. Understand character, string, and Boolean data types in Java.

CLASS

A class is a user defined blueprint or prototype from which objects are created. It represents the set of properties or methods that are common to all objects of one.

Create java class

You can use the New Java Class wizard to create a Java class. The Java Class wizard can be invoked in different ways −

- By clicking on the File menu and selecting New → Class.
- By right clicking in the package explorer and selecting New → Class.
- By clicking on the class drop down button ( ) and selecting class ( ).

Before bringing up the New Java Class wizard, if possible, select the package in which the class is to be created so that the wizard can automatically fill in the package name for you.

USING THE NEW JAVA CLASS WIZARD

Once the java class wizard comes up −

- Ensure the source folder and package are correct.
- Enter the class name.
- Select the appropriate class modifier.
- Enter the super class name or click on the Browse button to search for an existing class.
- Click on the Add button to select the interfaces implemented by this class.

Exercise

A. Write a Java program that:
- Declares and initializes a char variable for gender (M or F)
- Declares and initializes a boolean variable for driving license status (true or false)
- Displays the following output using escape sequences:

Expected Output:

```
Output - SampleTest (run)
run:
Gender: M
Has driving license: true
BUILD SUCCESSFUL (total time: 0 seconds)
```

Program:

```java
public class Lab3 {
    public static void main(String[] args) {

        char gender = 'M';
        boolean hasDrivingLicense = true;

        System.out.print("Gender: " + gender + "\nHas driving license: " + hasDrivingLicense);
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 3> java Lab3
Gender: M
Has driving license: true
PS C:\Users\Ansharah\Desktop\Java\Lab 3>
```

B. Write a Java program that:
- Take two String variables for first name and last name as user input.
- Concatenates them to display the full name in the following format using escape sequences:

Expected Output:

```
sampletest.SampleTest > main >
Output - SampleTest (run)
run:
Full Name: "John Doe"
BUILD SUCCESSFUL (total time: 0 seconds)
```

Program:

```java
import java.util.Scanner;

public class Lab3 {
    Run | Debug
    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        System.out.print(s: "Enter first name: ");
        String firstName = input.nextLine();

        System.out.print(s: "Enter last name: ");
        String lastName = input.nextLine();

        System.out.print("Full Name: \"" + firstName + " " + lastName + "\"");
        input.close();
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 3> javac Lab3.java
PS C:\Users\Ansharah\Desktop\Java\Lab 3> java Lab3
Enter first name: Ansharah
Enter last name: Khan
Full Name: "Ansharah Khan"
PS C:\Users\Ansharah\Desktop\Java\Lab 3>
```

C. Write a Java program to create a store billing system that:

- Takes the names of 5 items and their respective prices as input (prices should be of type double).
- Calculates the total bill.
- Type casts the total bill into an int and displays both the original and type-casted bill.

Expected Output:

```
Enter the name of item 1: Apple
Enter the price of Apple: 2.5
Enter the name of item 2: Milk
Enter the price of Milk: 1.75
Enter the name of item 3: Bread
Enter the price of Bread: 1.25
Enter the name of item 4: Cheese
Enter the price of Cheese: 3.50
Enter the name of item 5: Juice
Enter the price of Juice: 2.99

Total Bill (Original): $11.99
Total Bill (Rounded): $11
```

Program:

```java
import java.util.Scanner;

public class Lab3 {
    Run | Debug
    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        System.out.print(s: "Enter the name of item 1: ");
        String item1 = input.nextLine();
        System.out.print("Enter the price of " + item1 + ": ");
        double price1 = input.nextDouble();
        input.nextLine();

        System.out.print(s: "Enter the name of item 2: ");
        String item2 = input.nextLine();
        System.out.print("Enter the price of " + item2 + ": ");
        double price2 = input.nextDouble();
        input.nextLine();

        System.out.print(s: "Enter the name of item 3: ");
        String item3 = input.nextLine();
        System.out.print("Enter the price of " + item3 + ": ");
        double price3 = input.nextDouble();
        input.nextLine();
```

```java
        System.out.print(s: "Enter the name of item 4: ");
        String item4 = input.nextLine();
        System.out.print("Enter the price of " + item4 + ": ");
        double price4 = input.nextDouble();
        input.nextLine();

        System.out.print(s: "Enter the name of item 5: ");
        String item5 = input.nextLine();
        System.out.print("Enter the price of " + item5 + ": ");
        double price5 = input.nextDouble();

        double totalBill = price1 + price2 + price3 + price4 + price5;
        int roundedBill = (int) totalBill;

        System.out.print("\nTotal Bill (Original): $" + totalBill + "\nTotal Bill (Rounded): $" + rou
        input.close();
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 3> javac Lab3.java
PS C:\Users\Ansharah\Desktop\Java\Lab 3> java Lab3
Enter the name of item 1: Apple
Enter the price of Apple: 45
Enter the name of item 2: Banana
Enter the price of Banana: 60
Enter the name of item 3: Eggs
Enter the price of Eggs: 90
Enter the name of item 4: Potato
Enter the price of Potato: 100
Enter the name of item 5: Cheese
Enter the price of Cheese: 88

Total Bill (Original): $383.0
Total Bill (Rounded): $383
PS C:\Users\Ansharah\Desktop\Java\Lab 3>
```

D. Write a Java program to create an employee salary management system that:

1. **Takes the following inputs from the user**:
   - Employee Name (String)
   - Employee ID (int)
   - Basic Salary (double)
   - Allowance (float)
   - Tax Rate (in percentage as int)
2. **Calculates the following**:
   - Gross Salary = Basic Salary + Allowance
   - Tax Amount = (Gross Salary * Tax Rate) / 100
   - Net Salary = Gross Salary - Tax Amount
3. **Type cast the following**:
   - Convert tax rate to ***double*** for calculation
   - Convert net salary to ***int*** for display purposes

Expected Output:

```
Employee Name: John Doe
Employee ID: 101
Basic Salary: $5000.50
Allowance: $200.75
Gross Salary: $5201.25
Tax Amount (15%): $780.18
Net Salary (Original): $4421.07
Net Salary (Rounded): $4421
```

Program:

```java
import java.util.Scanner;

public class Lab3 {
    Run | Debug
    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);


        System.out.print(s: "Enter Employee Name: ");
        String empName = input.nextLine();

        System.out.print(s: "Enter Employee ID: ");
        int empID = input.nextInt();

        System.out.print(s: "Enter Basic Salary: ");
        double basicSalary = input.nextDouble();

        System.out.print(s: "Enter Allowance: ");
        float allowance = input.nextFloat();

        System.out.print(s: "Enter Tax Rate (%): ");
        int taxRate = input.nextInt();

        // Calculations
        double grossSalary = basicSalary + allowance;
```

```java
    double taxRateCasted = (double) taxRate;
    double taxAmount = (grossSalary * taxRateCasted) / 100;

    double netSalaryOriginal = grossSalary - taxAmount;
    int netSalaryRounded = (int) netSalaryOriginal;

    // Display Output
    System.out.print("\nEmployee Name: " + empName +
                     "\nEmployee ID: " + empID +
                     "\nBasic Salary: $" + basicSalary +
                     "\nAllowance: $" + allowance +
                     "\nGross Salary: $" + grossSalary +
                     "\nTax Amount (" + taxRate + "%): $" + taxAmount +
                     "\nNet Salary (Original): $" + netSalaryOriginal +
                     "\nNet Salary (Rounded): $" + netSalaryRounded);

    input.close();
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 3> java Lab3
Enter Employee Name: Ansharah
Enter Employee ID: 2245
Enter Basic Salary: 70000
Enter Allowance: 10000
Enter Tax Rate (%): 15

Employee Name: Ansharah
Employee ID: 2245
Basic Salary: $70000.0
Allowance: $10000.0
Gross Salary: $80000.0
Tax Amount (15%): $12000.0
Net Salary (Original): $68000.0
Net Salary (Rounded): $68000
PS C:\Users\Ansharah\Desktop\Java\Lab 3>
```

E. Write a Java program that:

1. **Takes the following inputs from the user**:
   - Length (double)
   - Width (double)

2. **Calculates:**
   - Area = Length × Width
   - Perimeter = 2 × (Length + Width)

3. Type cast the **area** to an **int** and display it.

Expected Output:

```
sampletest.SampleTest > main >
Output - SampleTest (run)
run:
Enter length: 5.75
Enter width: 3.5

--- Rectangle Details ---
Length: 5.75
Width: 3.5
Area (Original): 20.125
Area (Rounded): 20
Perimeter: 18.5
BUILD SUCCESSFUL (total time: 12 seconds)
```

Program:

```java
import java.util.Scanner;
public class SampleTest {
    Run | Debug
    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        System.out.print(s: "Enter length: ");
        double length = input.nextDouble();

        System.out.print(s: "Enter width: ");
        double width = input.nextDouble();

        // Calculations
        double area = length * width;
        double perimeter = 2 * (length + width);

        // Type cast area to int
        int areaRounded = (int) area;

        System.out.print("\n--- Rectangle Details ---" +
                         "\nLength: " + length +
                         "\nWidth: " + width +
                         "\nArea (Original): " + area +
                         "\nArea (Rounded): " + areaRounded +
                         "\nPerimeter: " + perimeter);
        input.close();
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 3> javac SampleTest.java
PS C:\Users\Ansharah\Desktop\Java\Lab 3> java SampleTest
Enter length: 6.7
Enter width: 4.5

--- Rectangle Details ---
Length: 6.7
Width: 4.5
Area (Original): 30.150000000000002
Area (Rounded): 30
Perimeter: 22.4
PS C:\Users\Ansharah\Desktop\Java\Lab 3>
```